



**Address:- office no. 126, Raut Arcade, opp.
Subway near Railway Station Virar East.**

Contact No. 9834617404

CSS

CSS tutorial or CSS 3 tutorial provides basic and advanced concepts of CSS technology. Our CSS tutorial is developed for beginners and professionals. The major points of CSS are given below:

CSS stands for Cascading Style Sheet.

CSS is used to design HTML tags.

CSS is a widely used language on the web.

HTML, CSS and JavaScript are used for web designing. It helps the web designers to apply style on HTML tags.

CSS Selector

CSS selectors are used to select the content you want to style. Selectors are the part of CSS rule set. CSS selectors select HTML elements according to its id, class, type, attribute etc.

There are several different types of selectors in CSS.

CSS Element Selector

CSS Id Selector

CSS Class Selector

CSS Universal Selector

CSS Group Selector

CSS Element Selector

The element selector selects the HTML element by name.

CSS Id Selector

The id selector selects the id attribute of an HTML element to select a specific element. An id is always unique within the page so it is chosen to select a single, unique element.

CSS Class Selector

The class selector selects HTML elements with a specific class attribute. It is used with a period character . (full stop symbol) followed by the class name.

CSS Class Selector for specific element

If you want to specify that only one specific HTML element should be affected then you should use the element name with class selector.

CSS Universal Selector

The universal selector is used as a wildcard character. It selects all the elements on the pages

CSS Group Selector

The grouping selector is used to select all the elements with the same style definitions.

Grouping selector is used to minimize the code. Commas are used to separate each selector in grouping.

How to add CSS

CSS is added to HTML pages to format the document according to information in the style sheet. There are three ways to insert CSS in HTML documents.

Inline CSS

Internal CSS

External CSS

CSS Comments

CSS comments are generally written to explain your code. It is very helpful for the users who reads your code so that they can easily understand the code.

CSS Background

CSS background property is used to define the background effects on element. There are 5 CSS background properties that affects the HTML elements:

background-color

background-image

background-repeat

background-attachment

background-position

CSS background-color

The background-color property is used to specify the background color of the element.

background-color: #b0d4de;

CSS background-image

The background-image property is used to set an image as a background of an element. By default the image covers the entire element. You can set the background image for a page like this.

background-image: url("paper1.gif");

CSS background-repeat

By default, the background-image property repeats the background image horizontally and vertically. Some images are repeated only horizontally or vertically

background-repeat: repeat-x

background-repeat: repeat-y;

CSS background-attachment

The background-attachment property is used to specify if the background image is fixed or scroll with the rest of the page in browser window. If you set fixed the background image then the image will not move during scrolling in the browser.

background-attachment: fixed;

CSS background-position

The background-position property is used to define the initial position of the background image. By default, the background image is placed on the top-left of the webpage.

You can set the following positions:

center

top

bottom

left

right

background: white url('good-morning.jpg');

background-repeat: no-repeat;

background-attachment: fixed;

background-position: center;

CSS Border

The CSS border is a shorthand property used to set the border on an element.

The CSS border properties are used to specify the style, color and size of the border of an element.

The CSS border properties are given below

border-style

border-color

border-width

border-radius

CSS border-style

The Border style property is used to specify the border type which you want to display on the web page.

none

It doesn't define any border.

dotted

It is used to define a dotted border.

dashed

It is used to define a dashed border.

solid

It is used to define a solid border.

double

It defines two borders with the same border-width value.

groove

It defines a 3d grooved border. effect is generated according to border-color value.

ridge

It defines a 3d ridged border. effect is generated according to border-color value.

inset

It defines a 3d inset border. effect is generated according to border-color value.

outset

It defines a 3d outset border. effect is generated according to border-color value.

CSS border-width

The border-width property is used to set the border's width. It is set in pixels. You can also use the one of the three pre-defined values, thin, medium or thick to set the width of the border.

CSS border-color

There are three methods to set the color of the border.

CSS Display

CSS display is the most important property of CSS which is used to control the layout of the element. It specifies how the element is displayed.

CSS display values

There are following CSS display values which are commonly used.

display: inline;

display: inline-block;

display: block;

display: none;

CSS display inline

The inline element takes the required width only. It doesn't force the line break so the flow of text doesn't break in inline

CSS display inline-block

The CSS display inline-block element is very similar to inline element but the difference is that you are able to set the width and height.

CSS display block

The CSS display block element takes as much as horizontal space as they can. Means the block element takes the full available width. They make a line break before and after them.

CSS display none

The "none" value totally removes the element from the page. It will not take any space

CSS Cursor

It is used to define the type of mouse cursor when the mouse pointer is on the element. It allows us

to specify the cursor type, which will be displayed to the user. When a user hovers on the link, then by default, the cursor transforms into the hand from a pointer.

alias

It is used to display the indication of the cursor of something that is to be created.

auto

It is the default property in which the browser sets the cursor.

all-scroll

It indicates the scrolling.

col-resize

Using it, the cursor will represent that the column can be horizontally resized.

cell

The cursor will represent that a cell or the collection of cells is selected.

```
<div style = "cursor:copy">copy value</div>
```

```
<div style = "cursor:pointer">Pointer</div>
```

CSS Buttons

In HTML, we use the button tag to create a button, but by using CSS properties, we can style the buttons. Buttons help us to create user interaction and event processing. They are one of the widely used elements of web pages.

background-color

border style

border-radius property

box-shadow

box-shadow : 0 8px 16px 0 black,
0 6px 20px 0 rgba(0, 0, 0, 0.19);

padding

CSS Float

The CSS float property is a positioning property. It is used to push an element to the left or right, allowing other element to wrap around it. It is generally used with images and layouts.

none

It specifies that the element is not floated, and will be displayed just where it occurs in the text. this is a default value.

left

It is used to float the element to the left.

right

It is used to float the element to the right.

CSS Font

CSS Font property is used to control the look of texts. By the use of CSS font property you can change the text size, color, style and more. You have already studied how to make text bold or underlined. Here, you will also know how to resize your font using percentage.

These are some important font attributes:

CSS Font color: This property is used to change the color of the text. (standalone attribute)

CSS Font family: This property is used to change the face of the font.

CSS Font size: This property is used to increase or decrease the size of the font.

CSS Font style: This property is used to make the font bold, italic or oblique.

CSS Font variant: This property creates a small-caps effect.

CSS Font weight: This property is used to increase or decrease the boldness and lightness of the font.

CSS Font Size

CSS font size property is used to change the size of the font.

xx-small

used to display the extremely small text size.

x-small

used to display the extra small text size.

small

used to display small text size.

medium

used to display medium text size.

large

used to display large text size.

x-large used to display extra large text size.

CSS Font Style

CSS Font style property defines what type of font you want to display. It may be italic, oblique, or normal.

```
h2 { font-style: italic; }
```

```
h3 { font-style: oblique; }
```

```
h4 { font-style: normal; }
```

CSS Font Variant

CSS font variant property specifies how to set font variant of an element. It may be normal and small-caps.

```
p { font-variant: small-caps; }
```

```
h3 { font-variant: normal; }
```

CSS Line Height

The CSS line height property is used to define the minimal height of line boxes within the element. It

sets the differences between two lines of your content.

CSS Margin

CSS Margin property is used to define the space around elements. It is completely transparent and doesn't have any background color. It clears an area around the element.

Top, bottom, left and right margin can be changed independently using separate properties. You can also change all properties at once by using shorthand margin property.

CSS Opacity

The CSS opacity property is used to specify the transparency of an element. In simple word, you can say that it specifies the clarity of the image.

In technical terms, Opacity is defined as degree in which light is allowed to travel through an object.

How to apply CSS opacity setting

Opacity setting is applied uniformly across the entire object and the opacity value is defined in term of digital value less than 1. The lesser opacity value displays the greater opacity. Opacity is not inherited.

```
opacity: 0.4;
```

```
filter: alpha(opacity=40); /* For IE8 and earlier */
```

```

```

```
<p>Transparent Image</p>
```

```

```

CSS filter

CSS filters are used to set visual effects to text, images, and other aspects of a webpage. The CSS filter property allows us to access the effects such as color or blur, shifting on the rendering of an element before the element gets displayed.

brightness()

As its name implies, it is used to set the brightness of an element. If the brightness is 0%, then it represents completely black, whereas 100% brightness represents the original one. It can also accept values above 100% that provide brighter results.

```
#img1 {  
    filter: brightness(130%);  
}
```

blur()

It is used to apply the blur effect to the element. If the blur value is not specified, then the value 0 is used as a default value. The parameter in blur() property does not accept the percentage values. A larger value of it creates more blur.

```
filter: blur(2px);
```

invert()

It is used to invert the samples in the input image. Its 100% value represents completely inverted, and 0% values leave the unchanged input. Negative values are not allowed in it.

`filter: invert(60);`

saturate()

It sets the saturation of an element. The 0% saturation represents the completely unsaturated element, whereas the 100% saturation represents the original one. The values greater than 100% are allowed that provides super-saturated results. We cannot use negative values with this property.

`filter: saturate(40);`

drop-shadow()

It applies the drop-shadow effect to the input image. The values it accepts are h-shadow, v-shadow, blur, spread, and color.

`filter: drop-shadow(10px 20px 30px yellow);`

grayscale()

It converts the input image into black and white. 0% grayscale represents the original one, whereas 100% represents completely grayscale. It converts the object colors into 256 shades of gray

`filter: grayscale(80%);`

CSS font-stretch property

The font-stretch property in CSS allows us to select a normal, expanded, or condensed face from the font's family. This property sets the text wider or narrower compare to the default width of the font. It will not work on any font but only works on the font-family that has a width-variant face.

font-stretch: normal | semi-condensed |

CSS Colors

The color property in CSS is used to set the color of HTML elements. Typically, this property is used to set the background color or the font color of an element.

RGB format.

RGBA format.

Hexadecimal notation.

HSL.

HSLA.

Built-in color.

RGB Format

RGB format is the short form of 'RED GREEN and BLUE' that is used for defining the color of an HTML element simply by specifying the values of R, G, B that are in the range of 0 to 255.

This property is not supported in all browsers; that's why it is not recommended to use it.

Syntax

```
color: rgb(R, G, B);
```

RGBA Format

It is almost similar to RGB format except that RGBA contains A (Alpha) that specifies the element's transparency. The value of alpha is in the range 0.0 to 1.0, in which 0.0 is for fully transparent, and 1.0 is for not transparent.

Syntax

```
color: rgba(R, G, B, A);
```

CSS hover

The :hover selector is for selecting the elements when we move the mouse on them. It is not only

limited to the links. We can use it on almost every HTML element. To style the link to unvisited pages, we can use the :link selector. To style the link for visited pages, we can use the :visited selector and to style the active links we can use the :active selector.

CSS background-attachment property

The background-attachment property is used to specify that the background image is fixed or scroll with the rest of the page in the browser window.

This property has three values scroll, fixed, and local. Its default value is scroll, which causes the element to not scroll with its content. The local value of this property causes the element to scroll with the content. If we set the value to fixed, the background image will not move during scrolling in the browser.

background-attachment: scroll | fixed | local |
initial | inherit;

CSS background-size property

The background-size CSS property is used to set the size of a background image of an element. The background image can be stretched or constrained to fit into the existing space. It allows us to control the scaling of the background image.

background-size: auto | length | cover | contain | initial | inherit;

CSS Line Height

The CSS line height property is used to define the minimal height of line boxes within the element. It sets the differences between two lines of your content.

line-height: 200%;

CSS Margin

CSS Margin property is used to define the space around elements. It is completely transparent and doesn't have any background color. It clears an area around the element.

Top, bottom, left and right margin can be changed independently using separate properties. You can also change all properties at once by using shorthand margin property.

CSS Opacity

The CSS opacity property is used to specify the transparency of an element. In simple word, you can say that it specifies the clarity of the image.

In technical terms, Opacity is defined as degree in which light is allowed to travel through an object.

How to apply CSS opacity setting

Opacity setting is applied uniformly across the entire object and the opacity value is defined in term of digital value less than 1. The lesser opacity value displays the greater opacity. Opacity is not inherited.

```
opacity: 0.4;
```

```
filter: alpha(opacity=40); /* For IE8 and earlier */
```

CSS Overflow

The CSS overflow property specifies how to handle the content when it overflows its block level container.

We know that every single element on a page is a rectangular box and the size, positioning and behavior of these boxes are controlled via CSS.

CSS Padding

CSS Padding property is used to define the space between the element content and the element border.

It is different from CSS margin in the way that CSS margin defines the space around elements. CSS padding is affected by the background colors. It clears an area around the content.

CSS Position

The CSS position property is used to set position for an element. it is also used to place an element

behind another and also useful for scripted animation effect.

You can position an element using the top, bottom, left and right properties. These properties can be used only after position property is set first. A position element's computed position property is relative, absolute, fixed or sticky.

Let's have a look at following CSS positioning:

CSS Static Positioning

CSS Fixed Positioning

CSS Relative Positioning

CSS Absolute Positioning

CSS Static Positioning

This is a by default position for HTML elements. It always positions an element according to the normal flow of the page. It is not affected by the top, bottom, left and right properties.

CSS Fixed Positioning

The fixed positioning property helps to put the text fixed on the browser. This fixed test is

positioned relative to the browser window, and doesn't move even you scroll the window.

CSS Relative Positioning

The relative positioning property is used to set the element relative to its normal position.

CSS- Absolute Positioning

The absolute positioning is used to position an element relative to the first parent element that has a position other than static. If no such element is found, the containing block is HTML.

CSS Vertical Align

The CSS vertical align property is used to define the vertical alignment of an inline or table-cell box. It is the one of the self-explanatory property of CSS. It is not very easy property for beginners.

It is applied to inline or inline-block elements.

top

It aligns the top of the element with the top of the tallest element on the line.

bottom

It aligns the bottom of the element with the lowest element on the line.

```
<p> This is an image with a text-top alignment.</p>
```

```
img.top {  
    vertical-align: text-top;  
}
```

CSS Width

The CSS width property is used to set the width of the content area of an element.

It does not include padding borders or margins. It sets width of the area inside the padding, border, and margin of the element.

CSS width values

auto

length

%

CSS Word Wrap

CSS word wrap property is used to break the long words and wrap onto the next line. This property is used to prevent overflow when an unbreakable string is too long to fit in the containing box.

Box-shadow CSS

It is used to add shadow-like effects around the frame of an element.

h-offset: It horizontally sets the shadow position. Its positive value will set the shadow to the right side of the box. Its negative value is used to set the shadow on the left side of the box.

v-offset: Unlike the h-offset, it is used to set the shadow position vertically. The positive value in it sets the shadow below the box, and the negative value sets the shadow above of the box.

blur: As its name implies, it is used to blur the box-shadow. This attribute is optional.

spread: It sets the shadow size. The spread size depends upon the spread value.

color: As its name implies, this attribute is used to set the color of the shadow. It is an optional attribute.

inset: Normally, the shadow generates outside of the box, but by using inset, the shadow can be created within the box

box-shadow: h-offset v-offset blur spread color
inset */ box-shadow: 5px 10px 10px 10px orange
inset;

CSS Text-shadow

As its name implies, this CSS property adds shadows to the text. It accepts the comma-separated list of shadows that applied to the text. It's default property is none. It applies one or more than one text-shadow effect on the element's text content.

h-shadow: It is the required value. It specifies the position of the horizontal shadow and allows negative values.

v-shadow: It is also the required value that specifies the position of the vertical shadow. It does not allow negative values.

blur-radius: It is the blur-radius, which is an optional value. Its default value is 0.

color: It is the color of the shadow and also an optional value.

```
text-shadow: -3px -3px 3px blue, 3px 3px 3px red;
```

Glow Effect

```
text-shadow: 0 0 .1em cyan;  
background-color: black;  
font-size:50px;  
text-align:center;  
}
```

CSS text-transform

This CSS property allows us to change the case of the text. It is used to control the text capitalization. This CSS property can be used to

make the appearance of text in all-lowercase or all-uppercase or can convert the first character of each word to uppercase.

Syntax

text-transform: capitalize | uppercase | lowercase
| none

capitalize

It transforms the first character of each word to uppercase. It will not capitalize the first letter after the number. It only affects the first letters of the words instead of changing the rest of the letters in the word.

text-transform: capitalize;

uppercase

As its name implies, it transforms all characters of the word into uppercase.

text-transform: uppercase;

lowercase

It transforms all characters of the word into **lowercase**.

text-transform: lowercase;

CSS Outline

CSS outline is just like CSS border property. It facilitates you to draw an extra border around an element to get visual attention

```
.box {  
    background-color: #eee;  
    outline: 3px solid red;  
    border: 3px solid lightgreen;  
    padding: 5px 10px;  
}
```

Difference between Border and Outline

At first glance, border and outline look similar, but there are some very important differences between them:

It is not possible to apply a different outline width, style and color for the four sides of an element while in border; you can apply the provided value for all four sides of an element.

CSS Visibility

The CSS visibility property is used to specify whether an element is visible or not.

CSS Visibility Parameters

visible: It is the by default value. It specifies that the element is visible.

hidden: It specifies that the element is invisible (but still takes up space).

CSS Animation

CSS Animation property is used to create animation on the webpage. It can be used as a replacement of animation created by Flash and JavaScript.

CSS3 @keyframes Rule

The animation is created in the @keyframe rule. It is used to control the intermediate steps in a CSS animation sequence.

CSS animation properties

@keyframes

It is used to specify the animation.

animation

This is a shorthand property, used for setting all the properties, except the animation-play-state and the animation-fill-mode property.

animation-delay

It specifies when the animation will start.

animation-direction

It specifies if or not the animation should play in reverse on alternate cycle.

animation-duration

It specifies the time duration taken by the animation to complete one cycle.

Keyframes timing function

For controlling the animation rate, we can use the following values.

linear: It means that the transition rate will be constant from start to end.

ease: It means that the animation starts slowly, and then after a time period, the speed increases, and before end speed will again slow down.

ease-in: It is similar to ease, but the animation ends quickly.

ease-out: It is also similar to ease, but the animation starts fast.

CSS radial-gradient() function

The radial-gradient() is an in-built CSS function which generates the smooth transitions between two or more than two colors. It sets the radial gradient as the background image. It can be a circular or elliptical shape.

background-image: radial-gradient(black, yellow, green, blue);

background-image: radial-gradient(circle, violet, indigo, blue, green, yellow, orange, red);

background-image: radial-gradient(farthest-side at left bottom, red, yellow, cyan);

background-image: radial-gradient(closest-corner , blue, yellow, green);

background-image: radial-gradient(closest-side , red, yellow, lightgreen);

CSS Gradient

CSS gradient is used to display smooth transition within two or more specified colors.

Why CSS Gradient

These are the following reasons to use CSS gradient.

You don't have to use images to display transition effects.

The download time and bandwidth usage can also be reduced.

CSS Linear Gradient

`gradient(red, green);`

CSS Linear Gradient: (Top to Bottom)

CSS Linear Gradient: (Left to Right)

`background: -webkit-linear-gradient(left, red, green)`

CSS Linear Gradient: (Diagonal)

`background: -webkit-linear-gradient(left top, red, green);`

CSS Radial Gradient

You must have to define at least two color stops to create a radial gradient. It is defined by its center.

background: -o-radial-gradient(blue, green, red);

background: -moz-radial-gradient(blue 5%, green 15%, red 60%);

CSS Transforms

CSS3 supports transform property. This transform property facilitates you to translate, rotate, scale, and skews elements.

Transformation is an effect that is used to change shape, size and position.

There are two type of transformation i.e. 2D and 3D transformation supported in CSS3.

CSS 2D Transforms

The CSS 2D transforms are used to re-change the structure of the element as translate, rotate, scale and skew etc.

Following is a list of 2D transforms methods:

translate(x,y): It is used to transform the element along X-axis and Y-axis.

translateX(n): It is used to transform the element along X-axis.

translateY(n): It is used to transform the element along Y-axis.

rotate(): It is used to rotate the element on the basis of an angle.

scale(x,y): It is used to change the width and height of an element.

scaleX(n): It is used to change the width of an element.

scaleY(n): It is used to change the height of an element.

skewX(): It specifies the skew transforms along with X-axis.

skewY(): It specifies the skew transforms along with Y-axis.

matrix(): It specifies matrix transforms.

The translate() method

The CSS translate() method is used to move an element from its current position according to the given parameters i.e. X-axis and Y-axis.

The rotate() method

The CSS rotate() method is used to rotate an element clockwise or anti-clockwise according to the given degree.

The scale() method

The CSS scale() method is used to increase or decrease the size of an element according to the given width and height.

The skew() method

The CSS skew() method is used to skew an element along with X-axis and Y-axis according to the given angle.

The matrix() method

The CSS `matrix()` method combines all the six 2D transform methods altogether. It allows you to rotate, scale, translate, and skew elements.

Syntax:

The parameters of matrix method:

`matrix(scaleX(),skewY(),skewX(),scaleY(),translateX(),translateY())`

CSS clearfix

A clear float (or clearfix) is a way for an element to fix or clear the child elements so that we do not require to add additional markup. It resolves the error which occurs when more than one floated elements are stacked next to each other.

Suppose if we set the position of a sidebar to the left of the main content block, but we get the elements collapse and overlap on each other. We can understand it as if a child element is floated and taller than its parent element; it will overflow outside of its container.

To overcome this, we can use the `overflow: auto;` property to the containing element.

`<!DOCTYPE html>`

```
<html>
<head>
<style>
div {
    border: 3px solid red;
    padding: 5px;
    background-color: pink;
    font-size: 20px;
}img{
    float: right;
    border: 3px solid blue;}
p{
    font-size: 20px;
    clear: right;
}
.jtp{
    overflow: auto;
}
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Without Clearfix</h1>
```

```
<div>
```

```

```

Welcome to the Alvin IT Solutions Here, you can find the best tutorials that are easy to read and help you to clear your concepts.

```
</div>
```

```
<p>Use the overflow:auto; CSS property</p>
```

```
<h1>With Clearfix</h1>
```

```
<div class="jtp">
```

```

```

Welcome to the javaTpoint.com. Here, you can find the best tutorials that are easy to read and help you to clear your concepts.

```
</div>
```

```
</body>
```

```
</html>
```

CSS icons

Icons can be defined as the images or symbols used in any computer interface refer to an element. It is a graphical representation of a file or program that helps the user to identify about the type of file quickly.

Using the icon library is the easiest way to add icons to our HTML page. It is possible to format the library icons by using CSS. We can customize the icons according to their color, shadow, size, etc.

```
<!DOCTYPE html>
<html>
<head>
<title>CSS Icons</title>
<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/4.7.0/css/font-awesome.min.css">
<style>
body{
text-align:center;
background-color:lightblue;
```

```
}  
.fa{  
color:red;  
font-size:50px;  
margin:10px;  
}  
</style>  
</head>  
<body style="text-align:center">  
<h1>Font Awesome Library</h1>  
  <i class="fa fa-cloud"></i>  
  <i class="fa fa-file"></i>  
  <i class="fa fa-heart"></i>  
  <i class="fa fa-bars"></i>  
  <i class="fa fa-car"></i>  
</body>  
</html>
```

CSS Transition

The CSS transitions are effects that are added to change the element gradually from one style to another, without using flash or JavaScript.

You should specify two things to create CSS transition.

The CSS property on which you want to add an effect.

The time duration of the effect.

CSS Flexbox

CSS3 Flexible boxes also known as CSS Flexbox, is a new layout mode in CSS3.

The CSS3 flexbox is used to make the elements behave predictably when they are used with different screen sizes and different display devices. It provides a more efficient way to layout, align and distribute space among items in the container.

It is mainly used to make CSS3 capable to change its item's width and height to best fit for all available spaces. It is preferred over block model.

The CSS3 flexbox contains flex containers and flex items.

Flex container:The flex container specifies the properties of the parent. It is declared by setting the display property of an element to either flex or inline-flex.

Flex items:The flex items specify properties of the children. There may be one or more flex items inside a flex container.

Flex Direction Property

The flex-direction property is used to set the direction of the flexible items inside the flex container. its default value is row (left-to-right, top-to-bottom).

The other possible values are:

row-reverse

column

column-reverse

flex-direction: row-reverse;

.....

Flexbox Justify-Content

The CSS3 justify-content property is used to define the alignment along the main axis. It is a sub-property of CSS3 Flexbox layout module.

It sets the Flexbox container's item horizontally when the items don't use all the available space on the main axis.

Its possible values are:

flex-start:It is the default value. It sets the items at the beginning of the container

flex-end:It sets the items at the end of the container.

Center:It sets the items at the center of the container.

space-between:It sets the items with space between the lines.

space-around:It sets the items with space before, between, and after the lines.

Flexbox align-items

The CSS3 flexbox align-items property is used to set the flexible container's items vertically align

when the items do not use all available space on the cross-axis.

Its possible values are:

stretch:It is the default value. It specifies that Items are stretched to fit the container.

flex-start:It sets the items at the top of the container.

flex-end:It sets the items at the bottom of the container.

center:It sets the items at the center of the container (vertically).

baseline:It sets the items at the baseline of the container.